INCH-POUND

MIL-I-82776(OS)
29 December 1989
SUPERSEDING
WS 21908A
9 December 1986
(See 6.5)

MILITARY SPECIFICATION

INSULATOR, EXIT CONE
IN-PROCESS REQUIREMENTS FOR

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification covers the in-process requirements for one type of exit cone insulator, referred to herein as the insulator (see 6.1). The in-process requirements cover insulator characteristics which must be verified prior to assembly and fabrication of the end product.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

TT-I-735 Isopropyl Alcohol

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426 appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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MILITARY

MIL-D-3464 Desiccants, Activated, Bagged, Packaging Use and Static Dehumidification

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR-43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of WR-43 are available from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 740 Methyl Ethyl Ketone (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

UNITED STATES PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia Absorbent Gauze

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 **In-Process.** The insulator shall conform to the requirements specified herein.

3.2 **Surface condition.** Defects in the surface conditions of the outside diametrical surfaces of the subassembly (which consists of the insert insulation, aft exit ring, retention ring and aft liner) and the outside diametrical and aft face surfaces of the completed assembly shall not exceed the limits specified in table I.

TABLE I. Surface limits.

Defect	Permissible condition extent
1. Blemishes (see 6.3.1)	Blemishes shall not exceed 1 inch long, 0.020 inch deep, or 0.050 inch wide. No more than two per part. (See figure 1.)
2. Delaminations (see 6.3.2)	None allowed.
3. Cracks (see 6.3.3)	None allowed.
4. Wrinkled areas (see 6.3.4)	None allowed.

3.3 **Physical properties.** Physical properties of the silica cloth phenolic overwrap shall be in accordance with table II.

TABLE II. Physical properties.

Property	Limits	
	Minimum	Maximum
Specific gravity, 23/23 degrees Celsius (°C)	1.67	1.79
Residual volatiles, percent (%)	0.00	2.00
Resin content, %	30.00	38.00
Compressive strength, pounds per square inch (psi), with ply	27,000	60,000

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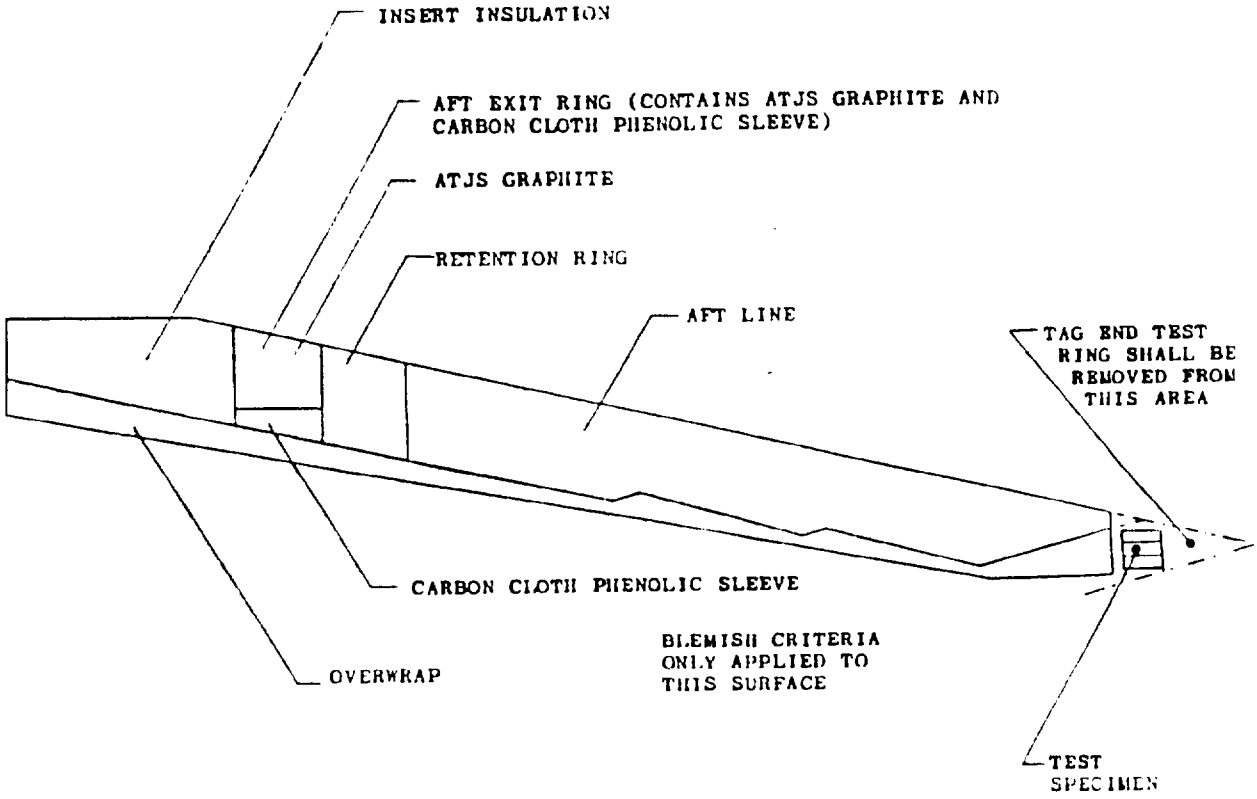


FIGURE 1. Exit cone insulator.

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3.4 **Interface bonds.** Separations and voids in the interface bonds shall not exceed the limits specified in table III.

TABLE III. Interface bond criteria.

Condition	Permissible Extent (Unbonds)
1. Aft exit ring to retention ring	None larger than 0.150 inch circumferentially by full bond width by full bond depth. No more than five in each bond line.
2. Retention ring to aft liner	None larger than 0.150 inch circumferentially by full bond width by full bond depth. No more than five in each bond line.
3. Carbon cloth phenolic sleeve to insert insulation	None larger than 0.150 inch circumferentially by full bond width by full bond depth. No more than five in each bond line.

3.5 **Workmanship.** The insulator shall be free from surface contamination, grease, oil, foreign material, and other defects which could render the part unsuitable for the intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an

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acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table IV.

TABLE IV. Quality conformance inspection.

Inspection	Requirement	Test Method
Visual examination	3.2, 3.4 & 3.5	4.3.1
Specific gravity test	3.3	4.3.2
Residual volatiles test	3.3	4.3.3
Resin content test	3.3	4.3.4
Compressive strength test	3.3	4.3.5

4.2.1 **Sampling for tests.**

4.2.1.1 **Lot.** For inspection purposes, a lot shall consist of all the insulators which are manufactured with the same lot of material and cured in the autoclave at the same time and subject to acceptance at the same time.

4.2.1.2 **Selection of samples.** Sample insulators shall be selected in accordance with table V for the physical properties tests specified in 4.3.2 through 4.3.5. Specimens for the tests shall be obtained as specified in 4.2.2.

TABLE V. Sampling for physical properties tests.

Number of Insulators in Lot	Number of Sample Insulators to be Selected for Testing
1 - 6	1
7 - 12	2
13 - 18	3
19 - 24	4

4.2.1.3 **Noncompliance of samples.** In the event that one or more of the samples fails to meet the requirements specified in 3.3, each insulator in the lot shall be subjected to the physical properties tests for the failed physical property, to determine compliance with requirements on an individual basis.

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4.2.2 **Test rings and specimens.** The test ring shall be removed from the area shown on figure 1 for each insulator to be subjected to the physical properties tests specified herein. The test ring shall be stored in a sealed bag and identified with the insulator's part number and serial number and the serial numbers of the other insulators in the lot (see 4.2.1). Test specimens shall be removed from each ring and prepared for testing as indicated in the applicable test methods.

4.2.3 **Classification of characteristics.** The characteristics verified by the tests and examination herein are classified as Major or Minor in accordance with WR-43. Tests that verify Major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the test. The visual examination herein is not annotated with a classification code or serial number and is classified Minor.

4.3 **Methods of inspection.** The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all inspections defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.3.1 **Visual examination.** Each insulator shall be visually examined to determine conformance to 3.2 and 3.5. Each insulator interface bond shall be visually examined to determine that separations and voids do not exceed the limits specified in 3.4. Noncompliance with any specified requirement shall constitute cause for rejection of the insulator. The following solvent wipe procedure shall be used as an examination aid in determining conformance to 3.2.

a. Wet surface with methyl ethyl ketone (MEK) conforming to ASTM D 740 or isopropyl alcohol conforming TT-I-735.

b. Wipe off excess solvent with type VIII absorbent gauze conforming to The United States Pharmacopeia.

c. Allow part to dry in air. Defect will appear as a well-defined wetted line or area when surrounding surface area appears dry.

4.3.2 **Specific gravity test.** Specific gravity for each insulator shall be determined in accordance with the following:

a. Remove three test specimens (0.30 $\pm$ 0.06 inch by 0.30 $\pm$ 0.06 inch by 0.60 $\pm$ 0.06 inch) 120 $\pm$ 24 degrees apart, from the test ring.

b. Clean each specimen using clean cloth or laboratory towel moistened with MEK or isopropyl alcohol, and allow to dry in air for a minimum of 20 minutes (see 4.3.1a).

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c. Weigh each specimen in air and record weight to nearest 0.1 milligram (mg).

d. Weigh each specimen in kerosene, and record weight to nearest 0.1 mg.

e. Calculate specific gravity of each specimen as follows:

$$\text{Specific gravity at } 23/23^{\circ}C = \frac{A \times C}{A - B}$$

Where: A = weight of specimen in air, g

B = weight of specimen in kerosene, g

C = specific gravity of kerosene at 23/23°C

f. Record specific gravity of each specimen to the nearest 0.01 unit.

g. Calculate the average specific gravity of the three specimens to the nearest 0.01 unit. Noncompliance of the average reading with the specific gravity requirements specified in 3.3 shall be cause for rejection of the insulator.

4.3.3 Residual volatiles test. Residual volatiles shall be determined in accordance with the following:

a. Remove three test specimens (0.30 ± 0.06 inch by 0.30 ± 0.06 inch by 0.60 ± 0.06 inch) 120 ± 24 degrees apart, from the test ring.

b. Clean each specimen, using clean cloth or laboratory towel moistened with MEK or isopropyl alcohol, and allow to dry in air for a minimum of 20 minutes (see 4.3.1a).

c. Pre-dry the specimens for minimum of 18 hours in desiccant conforming to MIL-D-3464.

d. Weigh each test specimen to nearest 0.001 g in a tared crucible which has been previously heated at 325 ± 10 degrees Fahrenheit (°F) for a minimum of 1 hour and cooled to room temperature in desiccator.

e. Place each crucible and specimen in an air circulating oven which has been preheated to, and maintained at, 325 ± 10°F.

f. Heat the specimens in the oven for 4 to 4.5 hours.

g. Remove crucibles and specimens from the oven, and cool in a desiccator to room temperature.

h. Weigh each crucible and contents to nearest 0.001 g.

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i. Calculate residual volatiles of each specimen as follows:

$$\text{Percent residual volatiles} = \frac{(A - B)}{A} \times 100$$

Where: A = sample weight before heating, g

B = sample weight after heating, g

j. Record average of three determinations to nearest 0.01 percent. Noncompliance of the average determination with the residual volatiles requirement specified in 3.3 shall be cause for rejection of the insulator.

k. Retain the three specimens for the resin content test (see 4.3.4).

4.3.4 **Resin content test.** (-M105) Resin content shall be determined in accordance with the following:

a. Obtain three test specimens from residual volatiles test (see 4.3.3) after volatile tests are completed.

b. Weigh each specimen to nearest 0.001 g in a tared crucible which has been previously fired at 1450 ± 50°F. Designate this weight as W1.

c. Place each specimen and crucible in muffle furnace.

d. Condition specimens and crucibles at 1450 ± 50°F for 4 hours minimum.

e. Cool specimens and crucibles in desiccator and reweigh to nearest 0.001 g. Designate this weight as W2.

f. Calculate percent resin solids for each specimen as follows:

$$\text{Percent resin solids} = \frac{W1 - W2}{W1} \times 100$$

Where: W1 = specimen weight after volatiles removal, g

W2 = specimen weight after volatiles removal and 4 hours at 1450 ± 50°F, g

g. Record resin content average for the three specimens to nearest 0.01 percent. Noncompliance of the average reading with the resin content requirement specified in 3.3 shall be cause for rejection of the insulator.

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4.3.5 **Compressive strength test.** (-M106) Compressive strength shall be determined in accordance with the following:

a. Remove three test specimens (0.300 ± 0.005 by 0.300 ± 0.005 by 0.600 ± 0.005 inch) 120 ± 24 degrees apart, from each test ring. Maximum surface roughness shall be 125 microinches root mean square (RMS). Machine the specimens such that length (0.600 inch reference) is tangent (within ± 18 degrees) to test ring curvature.

b. Measure the length, width, and height of all test specimens (to nearest 0.001 inch) with a calibrated micrometer. Record all readings on data sheet. Calculate and record the cross-section of each specimen.

c. Set the test specimen on load cell table of Instron tensile tester equipped with a Model CF load cell, with the length (0.600 inch reference) normal to the faces of the tester.

d. Depress the DOWN button to start compression. Cross-head speed should be 0.05 inch per minute.

e. When the test sample fails, push the RETURN button. This will return crosshead to ZERO position.

f. Repeat c through e for each other test specimen.

g. Calculate compressive strength of each specimen as follows:

$$\text{Compressive strength} = \frac{C}{A}$$

Where: C = maximum load, lbs

A = cross-sectional area, square inch

h. Record average compressive strength of the three specimens to nearest psi. Noncompliance of the average reading with the compressive strength requirement specified in 3.3 shall be cause for rejection of the insulator.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

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6.1 **Intended use.** The insulator specified herein is intended for use as a component in the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

6.3 **Definitions.**

6.3.1 **Blemishes.** Any superficial imperfection or marks or deformities such as scratches, crazing, gouges, cuts, or stains.

6.3.2 **Delamination.** A separation between fabric plies which has sharply defined boundaries that are parallel to the fabric plies.

6.3.3 **Crack.** A parting or separation which extends across the fabric plies.

6.3.4 **Wrinkled areas.** Distortions in fabric orientation normal to the plane of the fabric.

6.4 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.5 **Supersedure information.** MIL-I-82776 incorporates the following engineering change proposals (ECPs), notices of revision (NORs), specification change notices (SCNs):

ECPs	NORs	SCNs
----	----	SCN 2 (7/31/86)
MTA021 (11/24/86)	----	----
ARC001 (7/27/88)	ARC001.1 (7/27/88)	----
MTA101 (11/3/89)	MTA101.1 (11/7/89)	----

Preparing Activity:
NAVY-OS
(Project 1340-N785)

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-I-82776	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
INSULATOR, EXIT CONE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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